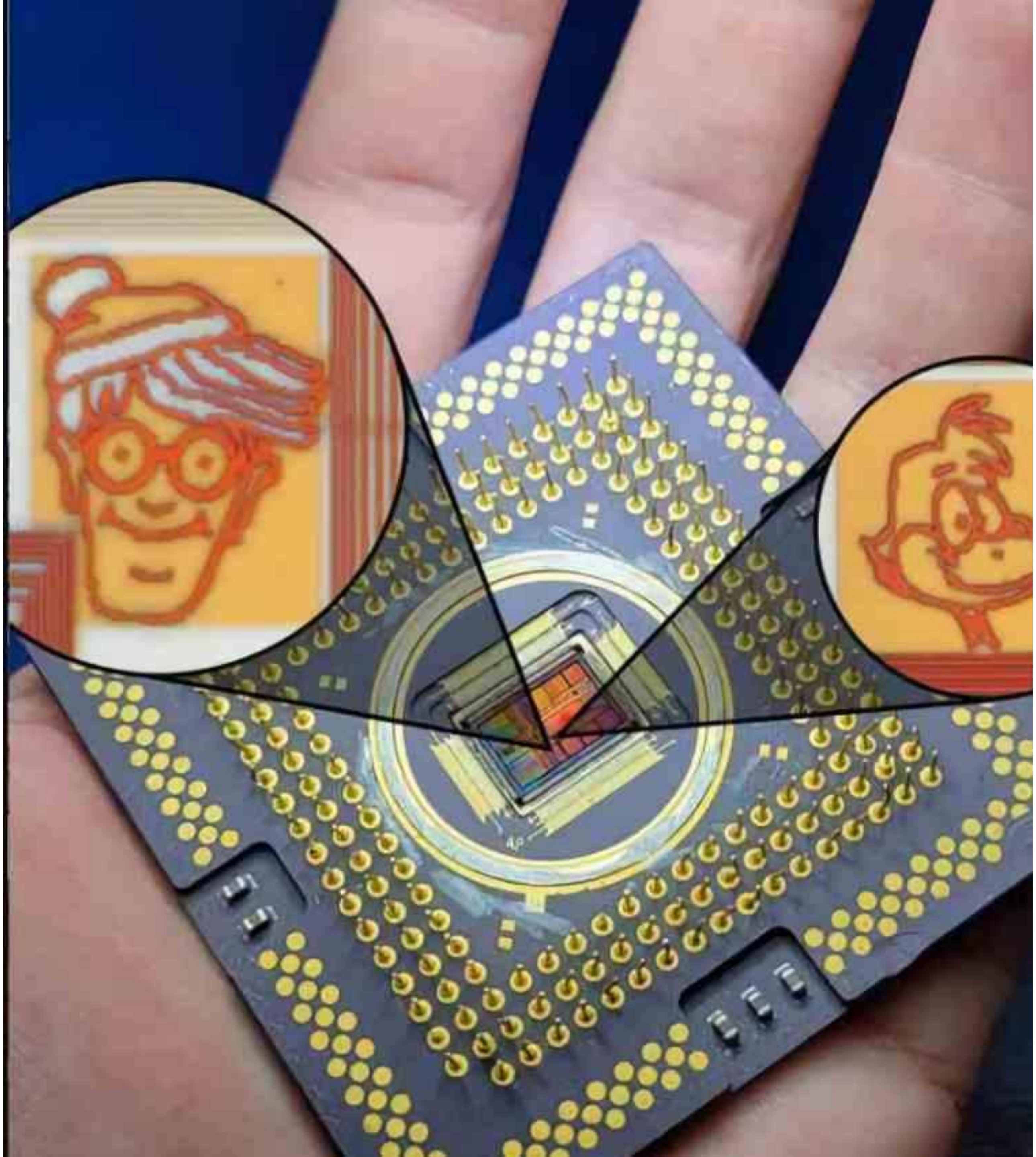
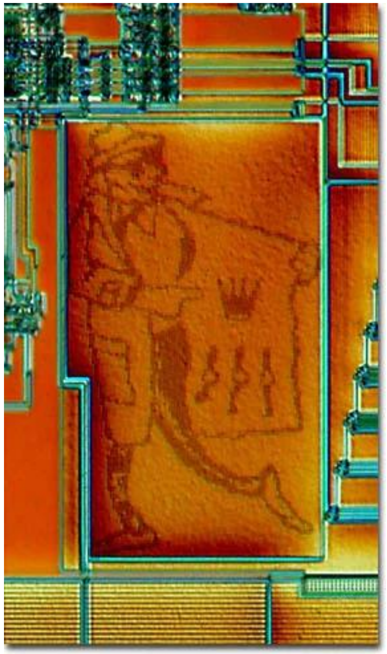
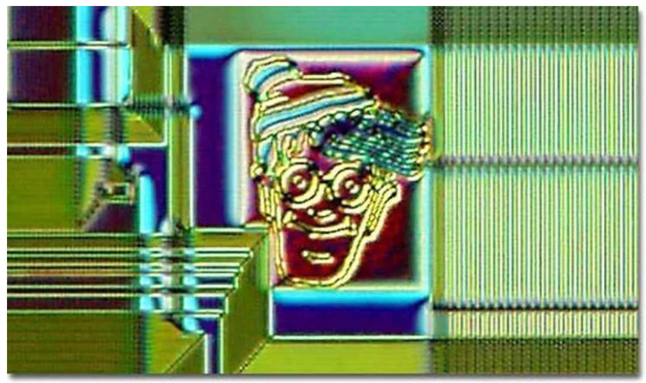
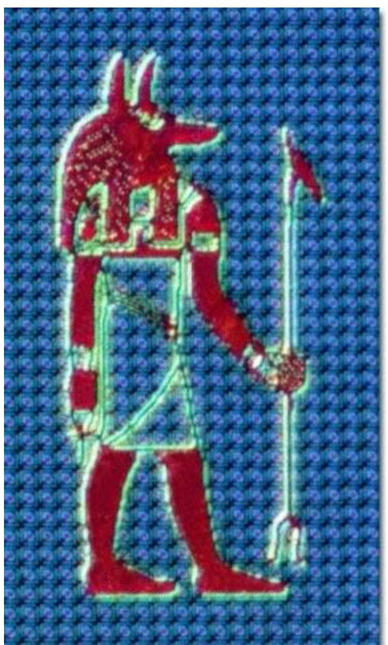

cadence
535-575 River Oaks

VIRTUOSO










$...
THE DREAM
IS ALWAYS
THE SAME

OPUS Environments

File	**display.drf**
Purpose	Specifies how you want your layers to appear on your monitor and in your plots.
User location	~/display.drf or ./display.drf
Sample location	cads01ac/sariti/test/default.drf
System default location	in some systems there is a common default

File	**cds.lib**
Purpose	Sets the paths to libraries and other cds.lib files.This file is used by the Library Browser, the File - Open command, and the File - New command.

Opening OPUS

cads01ac:sariti>mkdir test

- **Attention!**
- **Do not use space (or wildcards - !,?,*…) in file or directory names !**

cads01ac:sariti> cd test

cads01ac:sariti>icfb &

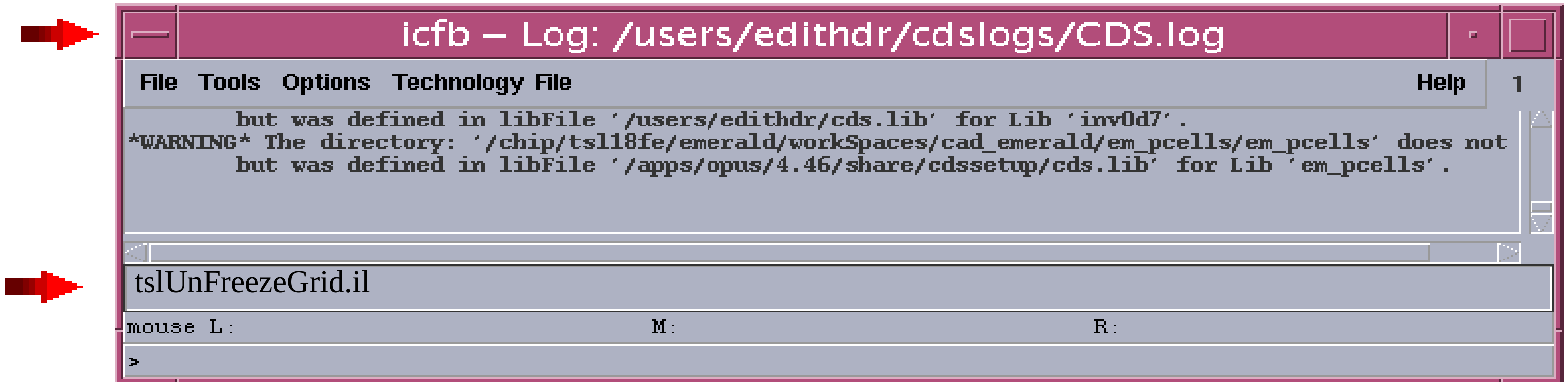
Command Interpreter Window Design Framework

Main window

You perform many tasks from the CIW:

- **Open windows**
- **Start tools**
- **View prompts, error messages, and informational messages**
- **Issue SKILL commands**
- **Make changes to your working environment**
- **Save window and form locations and session defaults**
- **Quit the design session**

Using the Command Interpreter Window



Title bar

The title bar shows the binary you are running. It also shows the path to the log file (CDS.log) that records the ongoing events of the design session. The content of the log file is displayed in the output area.

Input Line

The input line is where you type SKILL commands, coordinates of points, or any other keyboard input.

Using CIW

Output Area

The output area displays the commands you use and the responses the system makes during the design session.

Using CIW

These commands and responses are in SKILL **code. You can control what is displayed by changing the log filter.**

The output area normally displays only a few lines of code. To see more, use the scroll bars or resize the window. When you resize the CIW**, only the output area changes size.**

The information in the output area in log file, CDS.log**, that the system maintains on disk.**

To see the full record of activity, open CDS.log **from an** xterm **window.**

When you increase the size of the output area**, you see the current log file scroll past.**

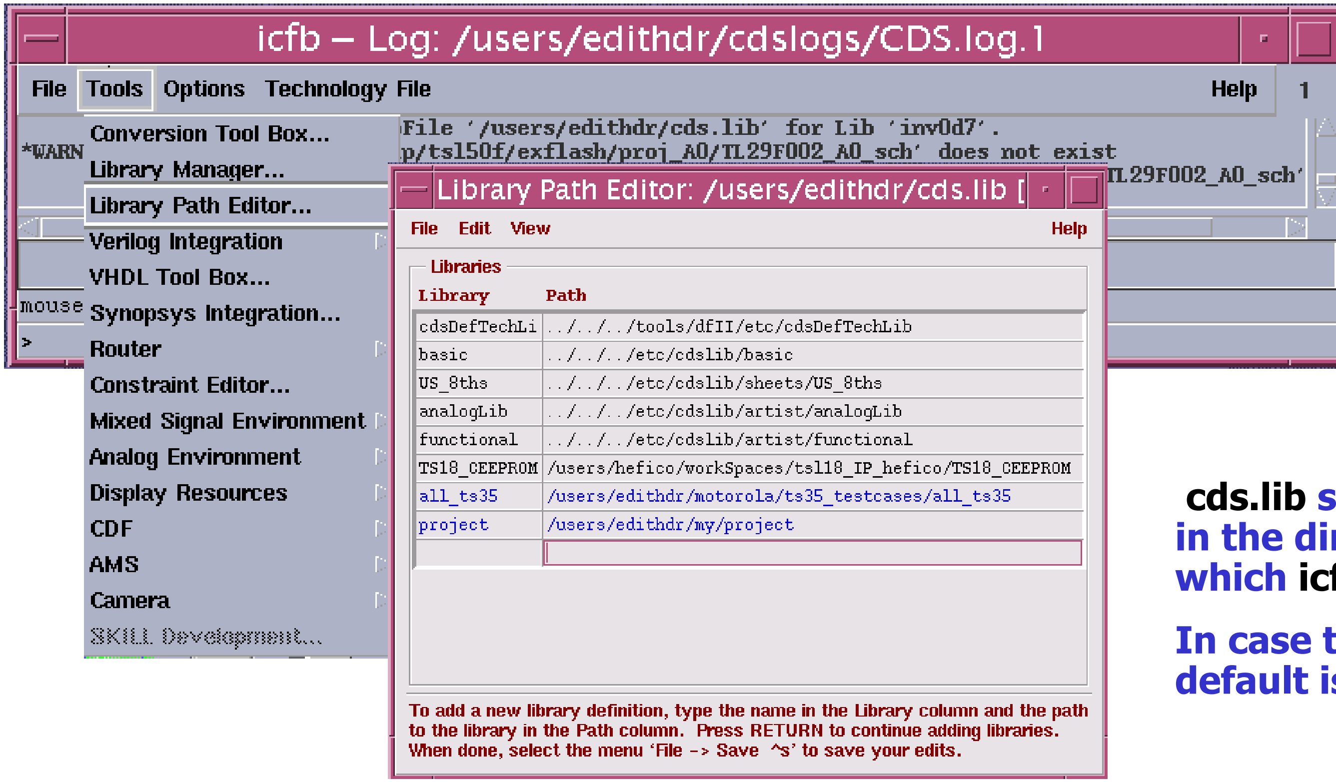
cds.lib

SOFTINCLUDE /users/nirgr/Pcellscds.lib

UNDEFINE TOSH_ZR78_GC

DEFINE MEI_STRIP1E /work/projects/MEI/libraries/MEI_STRIP1E

DEFINE analogLib /apps/cadence/4.46/tools.sun4v/dfII/etc/cdslib/artist/analogLib



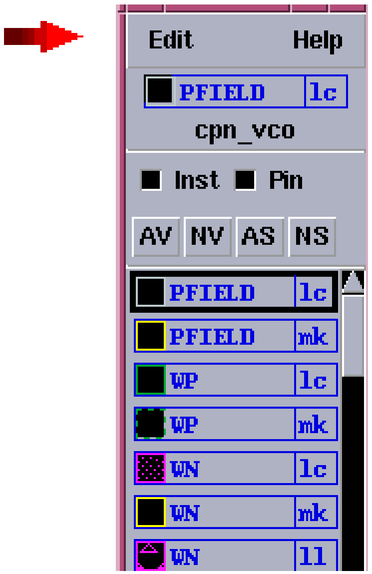
cds.lib should be present in the directory from which icfb is run.

In case there is no cds.lib default is created.

Using LSW

About the Layer Selection Window

The Layer Selection Window (LSW) lets you choose the design layer for each shape you create, make design layers visible or invisible, or make instances and pins selectable or unselectable.



Technology file

Each library must be associated with a technology file. The ***technology file*** defines:

• Design layers and their properties

• Physical design rules used for compaction, symbolic checking, and interactive verification

• Contacts used by the layout editor

• Devices and pins used by the Virtuoso compactor and layout synthesizer tools

• Application-specific rules that control how applications work

You can associate multiple libraries with a single technology file using the Attach To command.

Technology file example

You can see ASCII technology files for TS35 and TS50 in

/users/cad4fab/cad/tech/FAB1/TSL_A35.tf

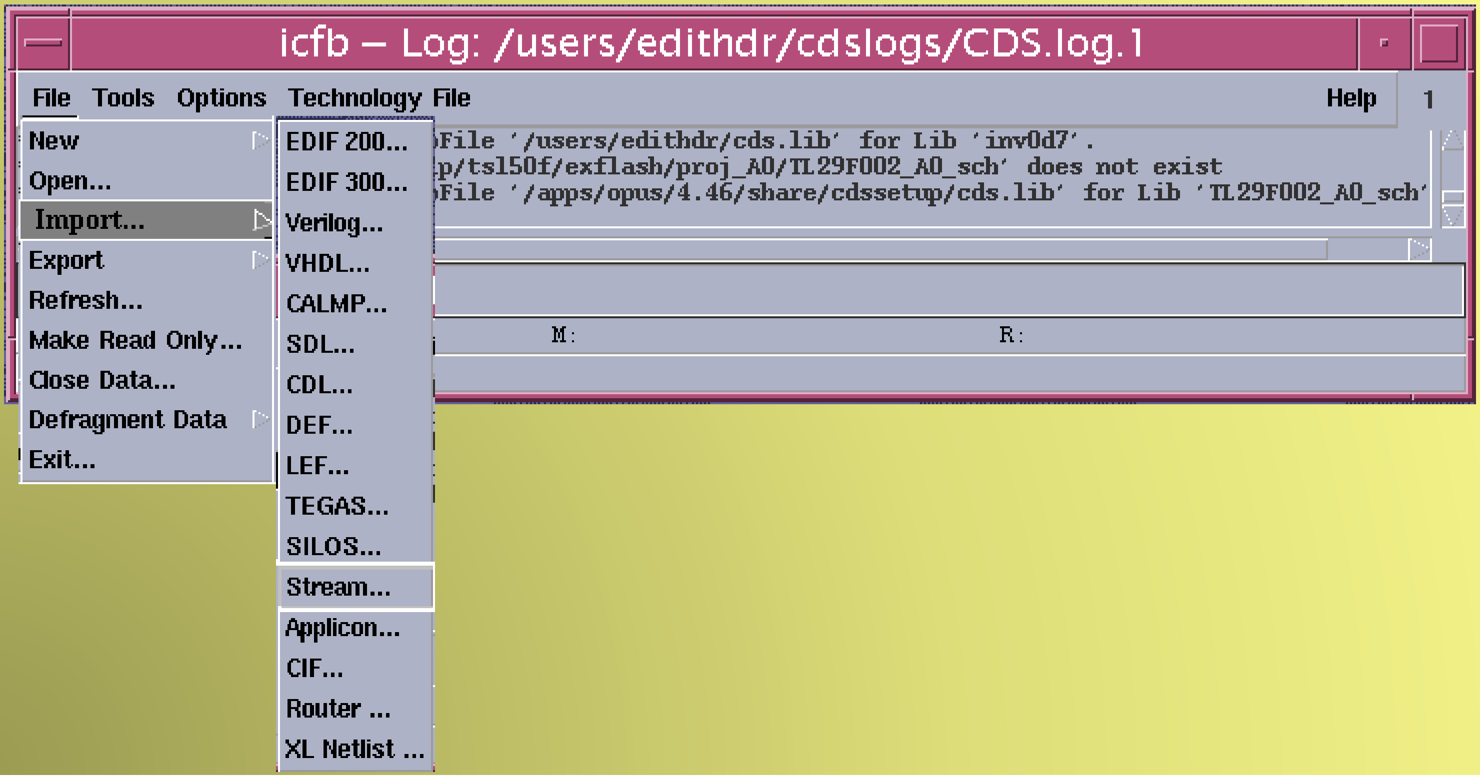
Technology file for TS018 you can see in

/users/cad4fab/cad/tech/018/TS18_SL_rev4_7_all.tf

Note: New versions of tech. Files will be created. Always check for latest one.

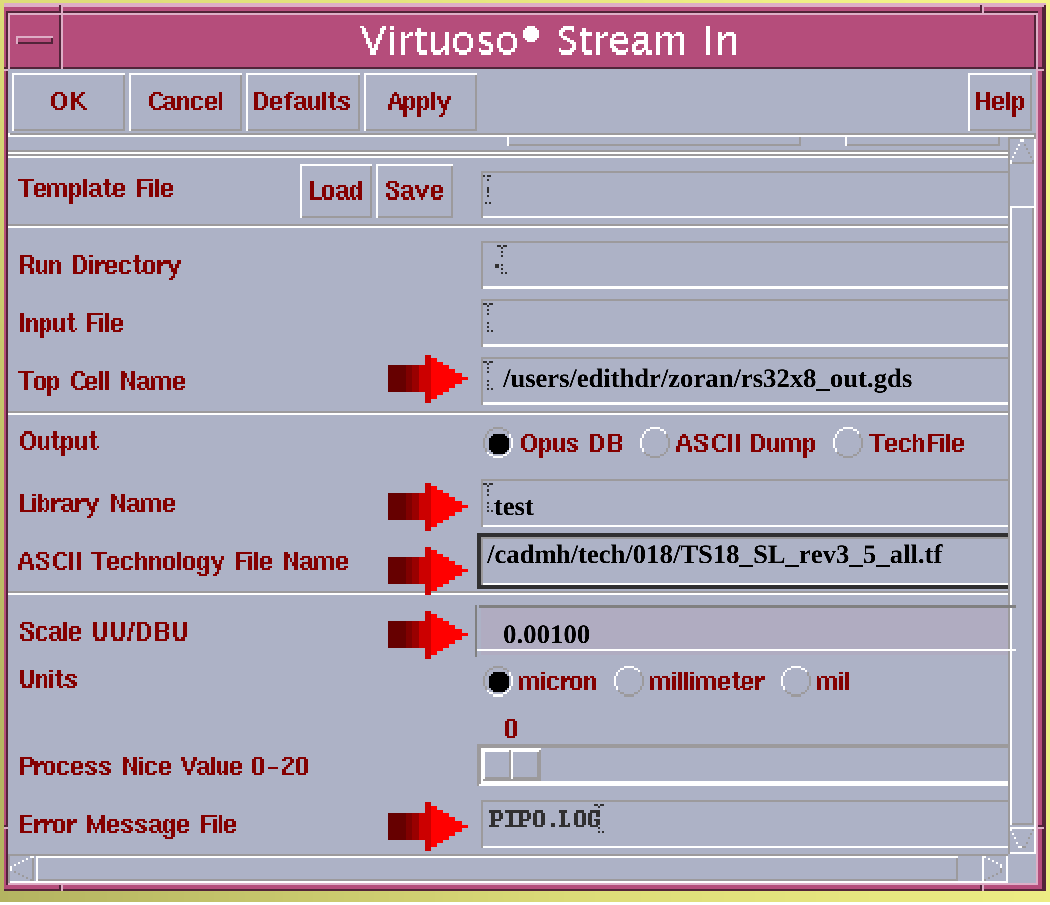
Stream In

- **Import -> Stream** translates Geometric Data Standard II (GDSII) Stream mask data from **Stream** format into Design Framework II (DFII) database format.

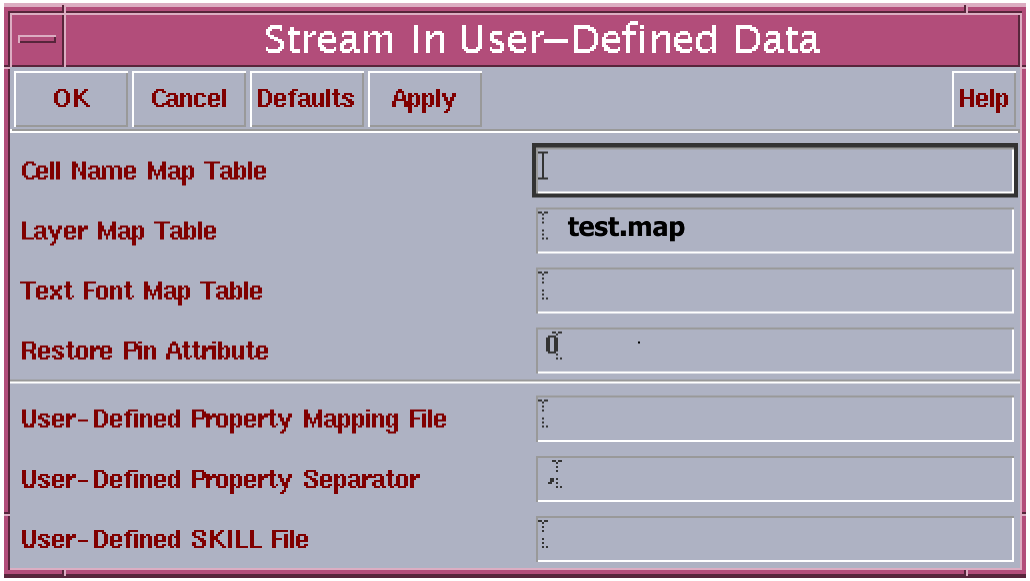


Stream In

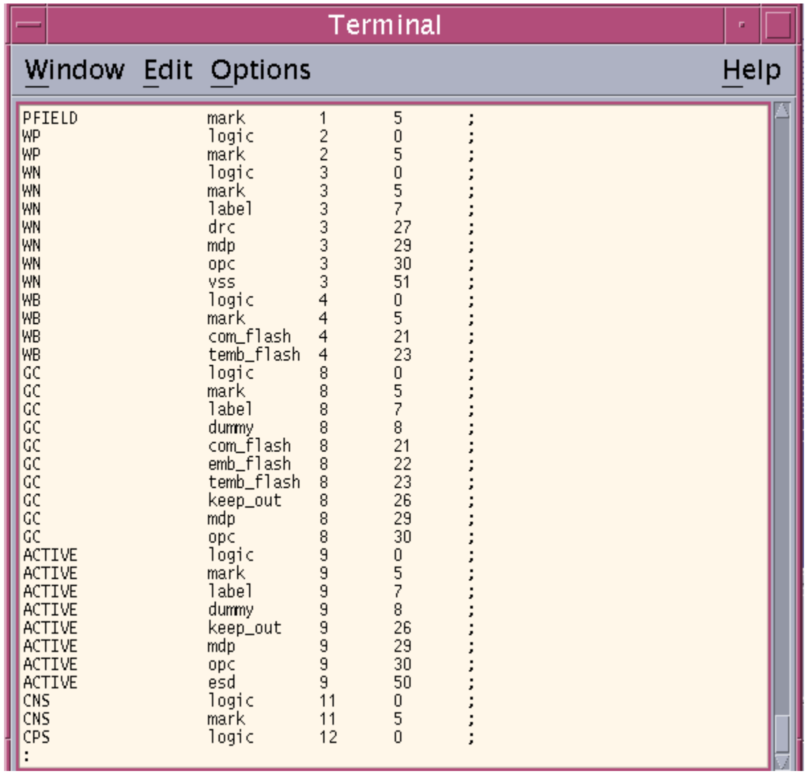
File ->Import ->StreamIn



Using map file



MAP FILE




```
PFIELD    mark        1    5    ;
WP        logic       2    0    ;
WP        mark        2    5    ;
WN        logic       3    0    ;
WN        mark        3    5    ;
WN        label       3    7    ;
WN        drc         3    27   ;
WN        mdp         3    29   ;
WN        opc         3    30   ;
WN        vss         3    51   ;
WB        logic       4    0    ;
WB        mark        4    5    ;
WB        com_flash   4    21   ;
WB        temb_flash  4    23   ;
GC        logic       8    0    ;
GC        mark        8    5    ;
GC        label       8    7    ;
GC        dummy       8    8    ;
GC        com_flash   8    21   ;
GC        emb_flash   8    22   ;
GC        temb_flash  8    23   ;
GC        keep_out    8    26   ;
GC        mdp         8    29   ;
GC        opc         8    30   ;
ACTIVE    logic       9    0    ;
ACTIVE    mark        9    5    ;
ACTIVE    label       9    7    ;
ACTIVE    dummy       9    8    ;
ACTIVE    keep_out    9    26   ;
ACTIVE    mdp         9    29   ;
ACTIVE    opc         9    30   ;
ACTIVE    esd         9    50   ;
CNS       logic       11   0    ;
CNS       mark        11   5    ;
CPS       logic       12   0    ;
:
```

Each line in the layer name map file contains four entries:

- **Design Framework II database layer name**
- **Design Framework II database layer purpose**
- **Stream layer number**
- **Stream data type.**

Notes: For FAB1 map.file is a must!

For FAB2 use map.file in case of non Tower layers only!

Create new library

For opening an existing library you need just click

To create a new library,

1. Choose File - New.

2. Choose Library.

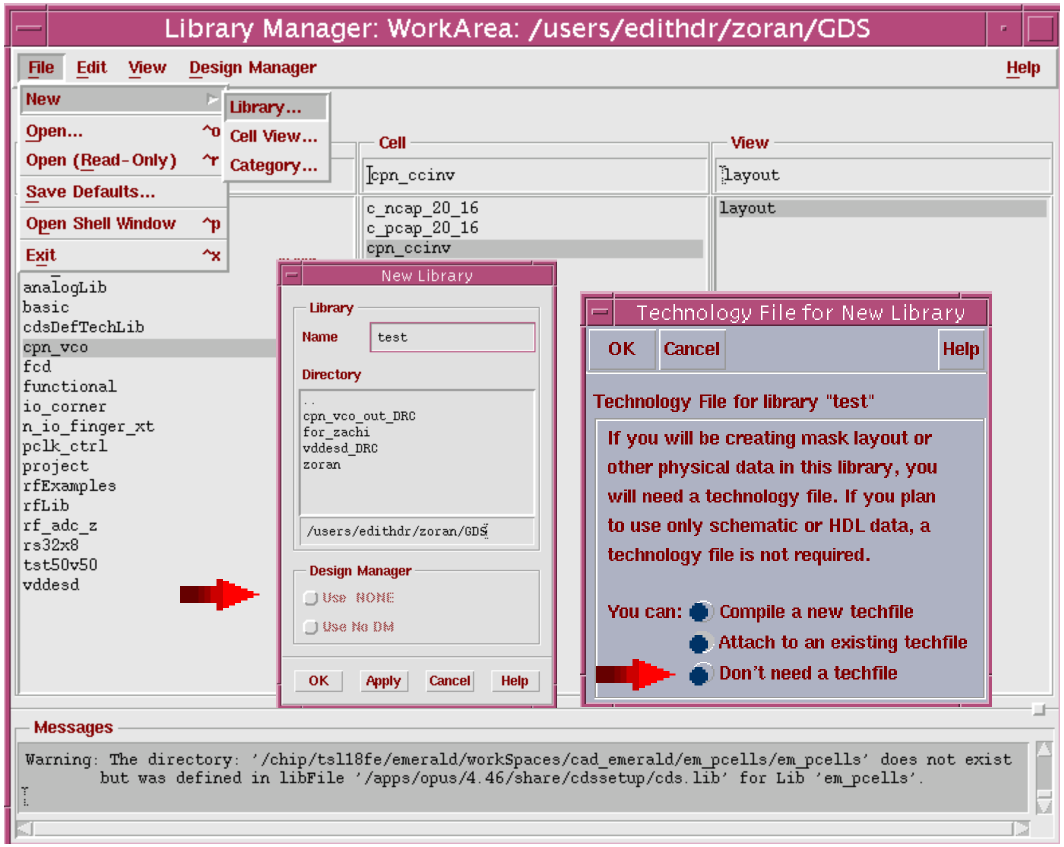
3. Type the name of the new library in the Library Name field.

4. Click OK.

The new library is created in your directory as an empty subdirectory. The New Library dialog box closes.

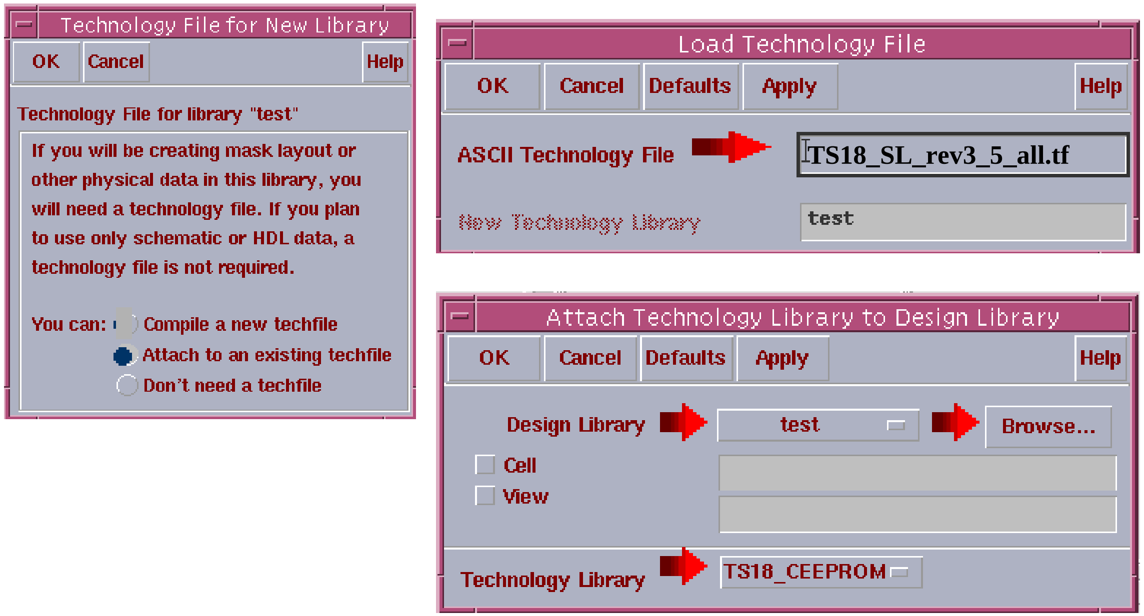
The New Library dialog box appears.

Create new library



For schematic only choose option "don't need a techfile"

Create new library



Those settings can be over written later using "technology file" option in CIW

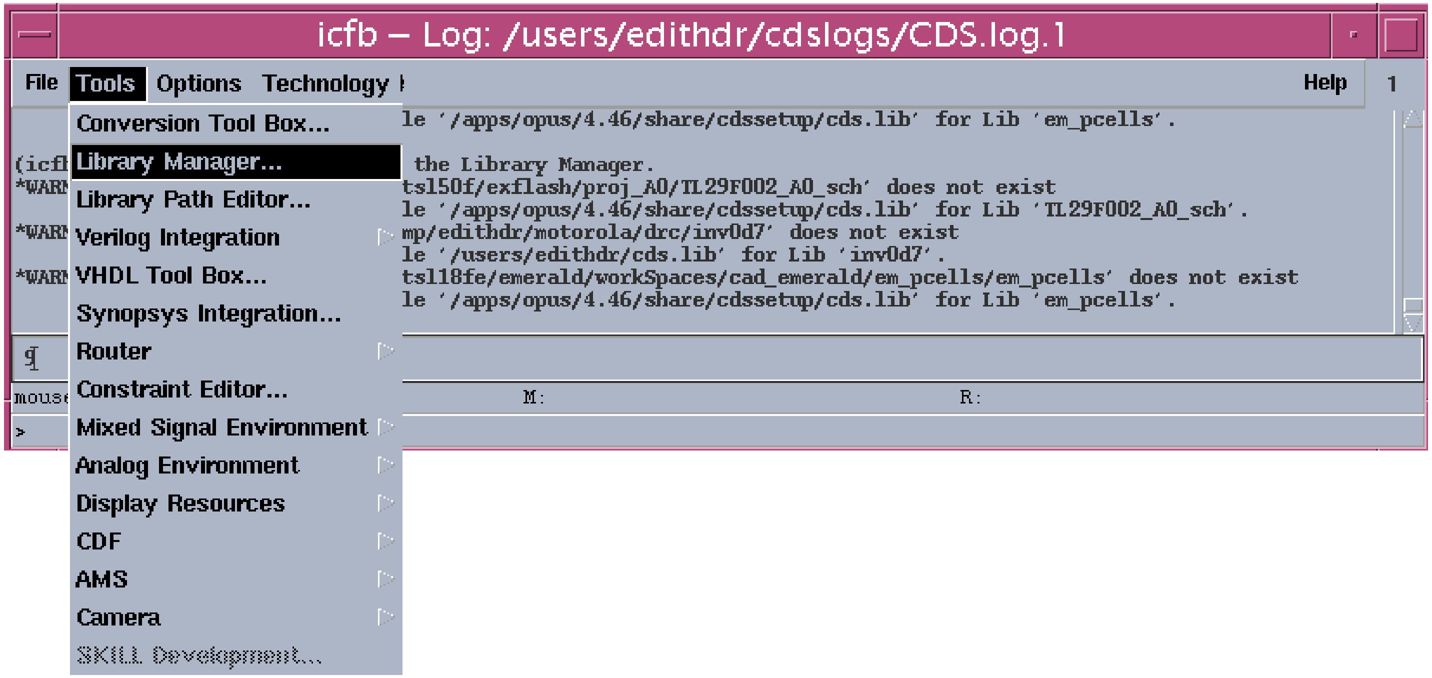
Open existing file

To open a view as read only from the library manager viewing area, do one of the following:

- **Click on the view and choose File -> Open (Read Only).**
- **Click the right mouse button on the view and choose Open (Read Only) in the pop-up menu.**

The view is opened but cannot be changed. If the library manager cannot open the item, an Open Error dialog box appears.

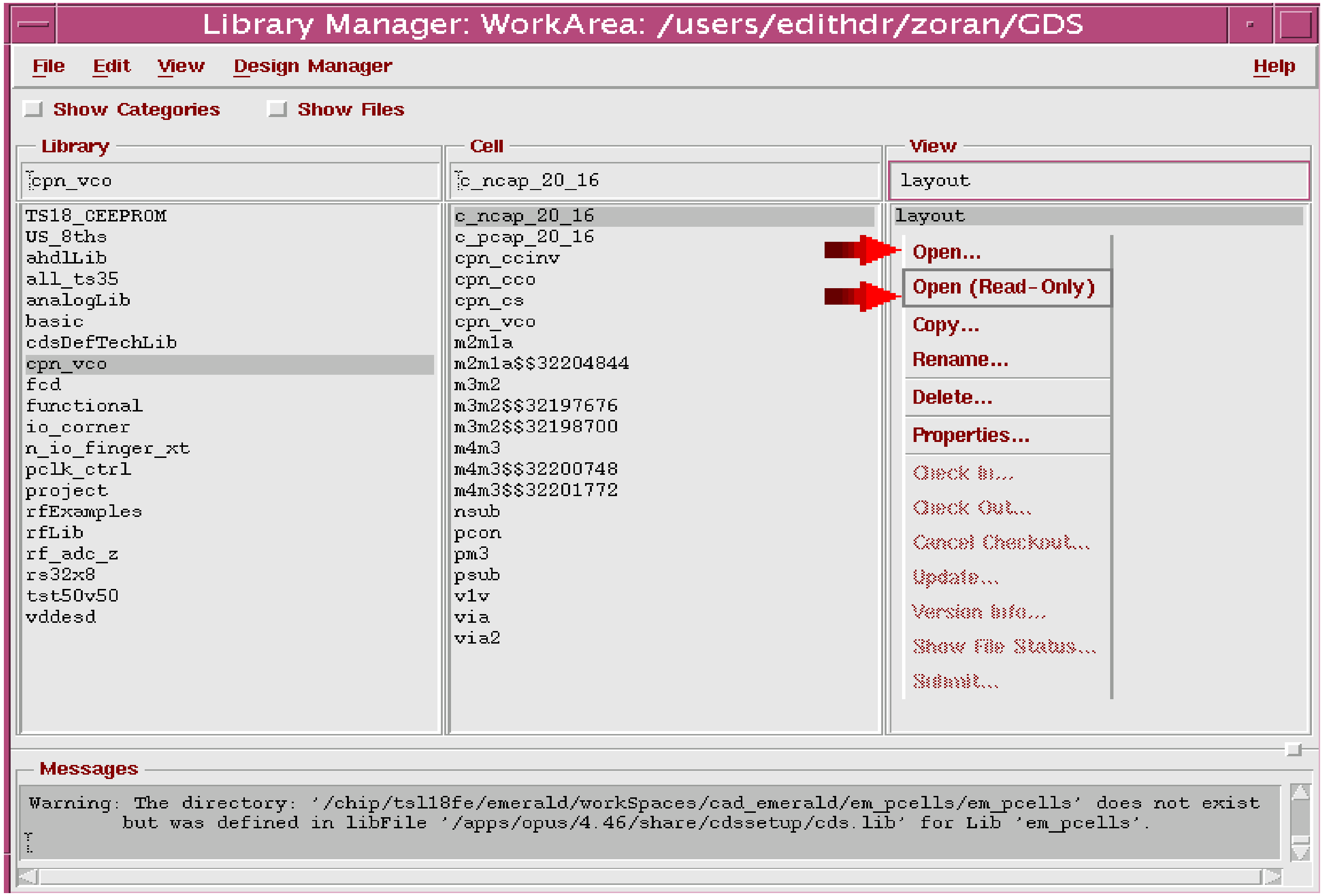
Open library manager



To open a view or file from the library manager viewing area, do one of the following:

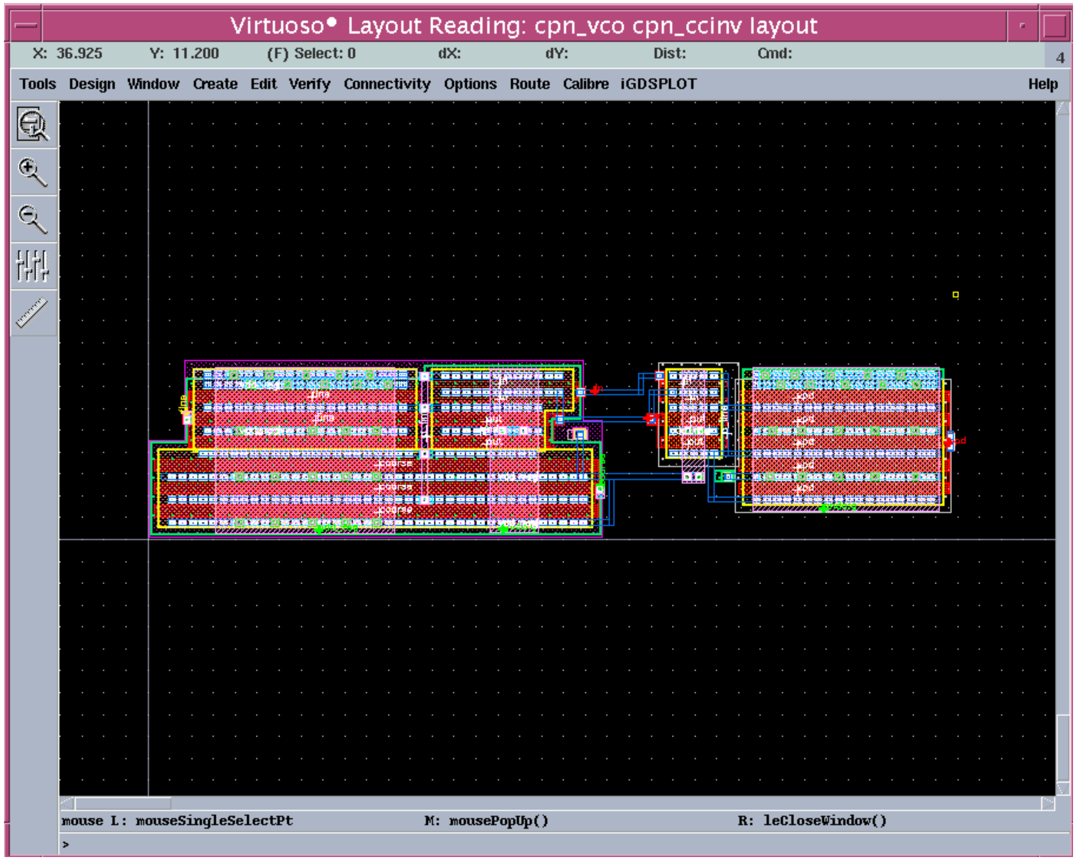
- **Double-click the item.**
- **Click on the item and choose File -> Open.**
- **Click the right mouse button on the item and choose Open in the pop-up menu.The item is opened. If the library manager cannot open the item, an Open Error dialog box appears. If the item is a text file, the library manager opens a text editor for the file.**

Opening layout

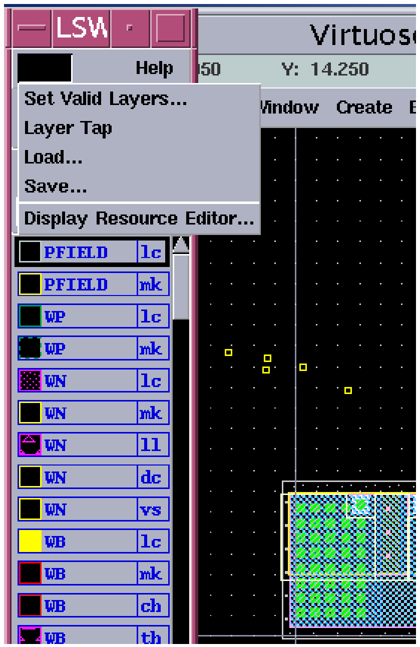


Note: Libraries are defined in cds.lib.

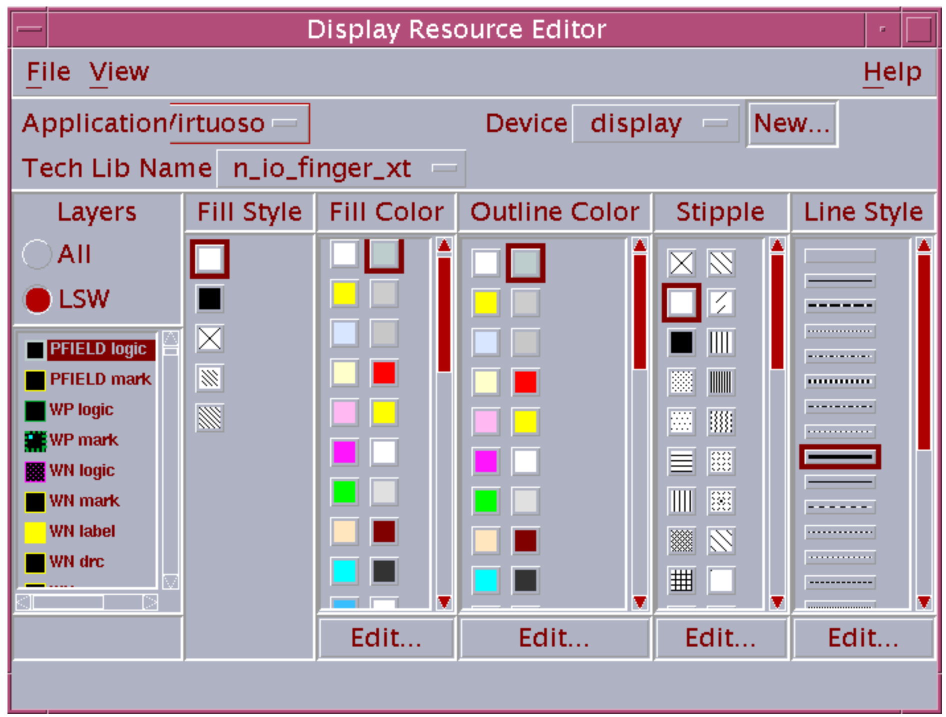
Opening layout



Change layer appearance



- Default Display.drf settings may be changed in Display Resource Editor
- Settings can be saved for future re-use.



Create new file

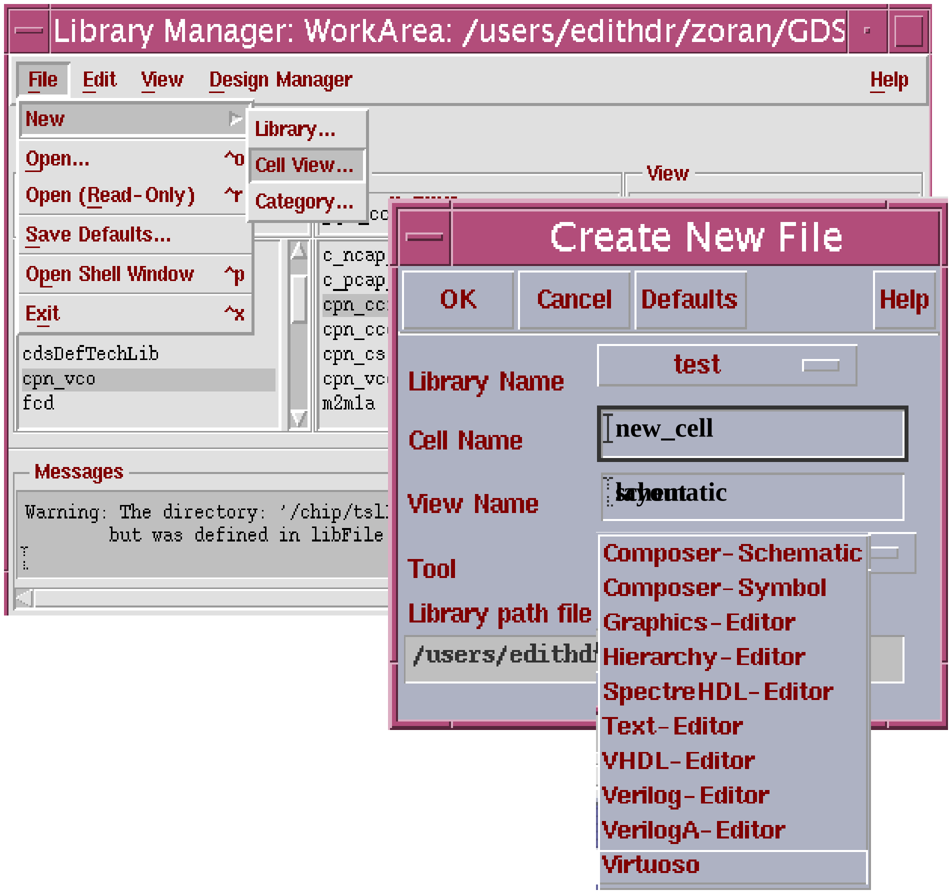
To create a new cell in a library,

1. **Select the library in which to place the new cell, or a cell in that library.**
2. **Choose File - New.**
3. **Choose Cell.**
4. **Verify the name of the library where the new cell will be placed.**
5. **Type the name of the new cell.**
6. **Type the name of the new view (layout, schematic, etc.**
7. **Type the name of the new view**
8. **Click OK.**

The new cell is created in the library as an empty subdirectory.

The New Cell dialog box closes

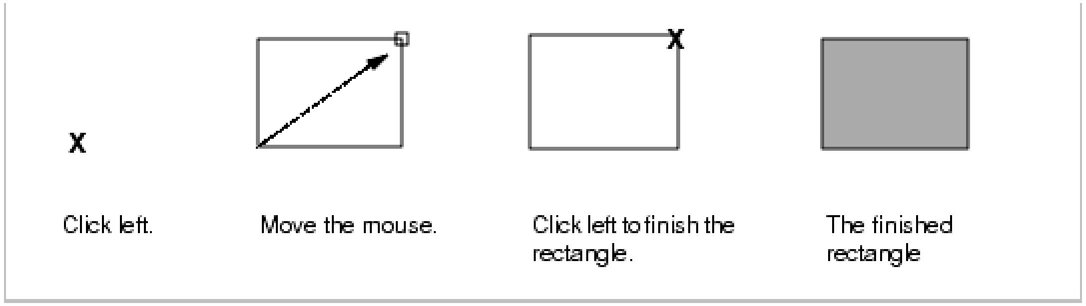
Create new layout



Basic layout editing

Create rectangle

1. In the Layer Selection Window, click on the layer you want.
2. Choose Create -> Rectangle [r].
3. Click to enter the first corner of the rectangle.
4. Click to enter the opposite corner of the rectangle.



Basic layout editing

- Create polygon

1.In the **LSW**, **click** on the layer you want.

2.Choose **Create - Polygon [Shift-p].**

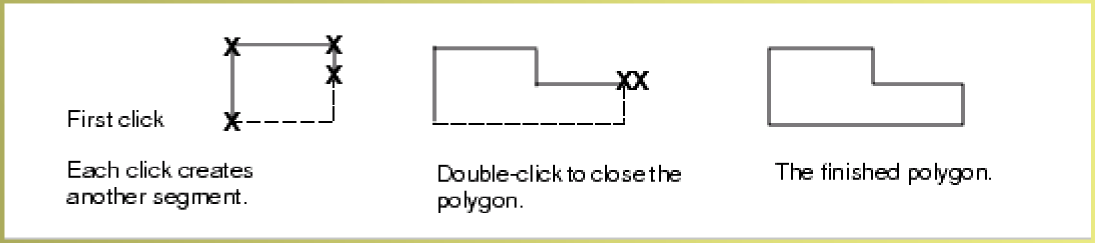
5.**Click** to enter the first point.

6.Move the cursor and **click** to enter the next point.

7.Continue to move the cursor and click to enter points.

As you enter each point, a dotted line shows how the layout editor will close the polygon if you stop at that point.

8.**Double-click** to finish the polygon.



Basic layout editing

- Create path

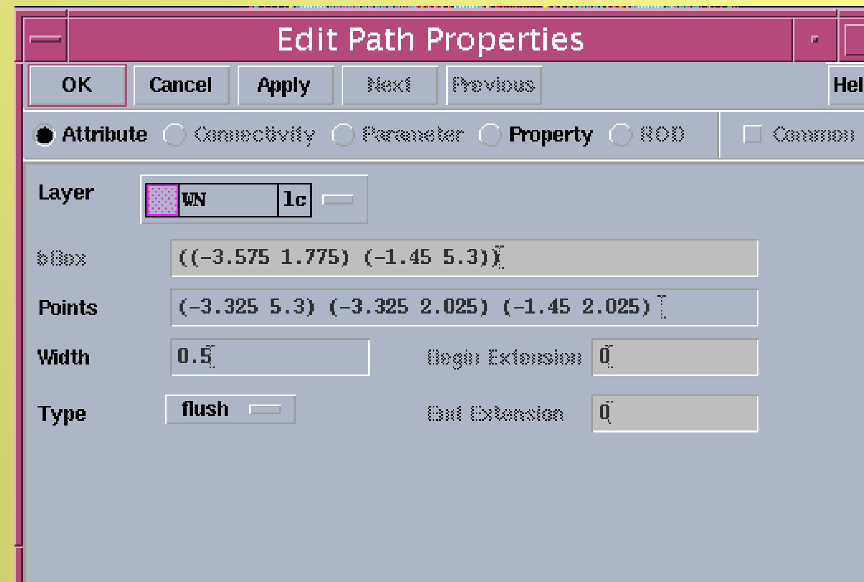
1. **In the LSW, click on the layer you want to use.**
2. **Choose Create -> Path.**
3. **Enter the first point by clicking in the cell view.**
4. **Move the cursor to the next point and click.**
5. **Continue to move the cursor and click to enter points.**
6. **Double-click to finish the path.**

Basic layout editing

Change path

You can change path in **Create** path form

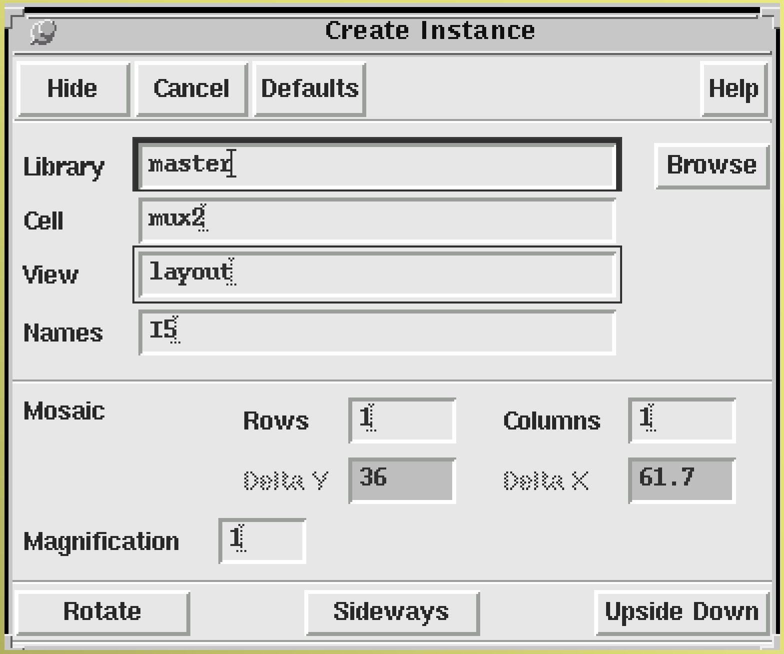
- **(F3)** before path drawing
- **Edit -> Properties** an existing path for change some options



Note: F3 button can be used in ALL layout editing operations!

Basic layout editing

- Create Instance Form



Basic layout editing

Library, **Cell**, and **View** set the library, cell, and view names of the master cell you want to place as an instance in this cell view.

Browse lets you select the library, cell, and view names by clicking on them in the browser.

Names sets the name assigned to this instance. You can type any name unique to the

cell view here or let the layout editor automatically assign instance names that begin with the letter I, followed by a number. You can enter multiple names

(separated by a space) to place several instances of the same cell.

Mosaic

Rows and Columns set the number of rows and columns in an array of instances.

Delta X and Delta Y set the spacing between rows and columns in an array of instances.

Magnification enlarges or reduces the size of the cell instance.

Rotate turns the instance 90 degrees counterclockwise.

Sideways mirrors the instance along the X axis.

Upside Down mirrors the instance along the Y axis.

Basic layout editing

To place a cell inside the current cell view,

1. Choose **Create -> Instance.**

   The **Create Instance form** appears.

2. Fill in the **Library**, **Cell**, and **View** fields.
3. Move the cursor into the cell view.

   An outline of the cell you want to place follows the cursor.

4. **Click** where you want to place the instance.

To rotate instances as you place them,

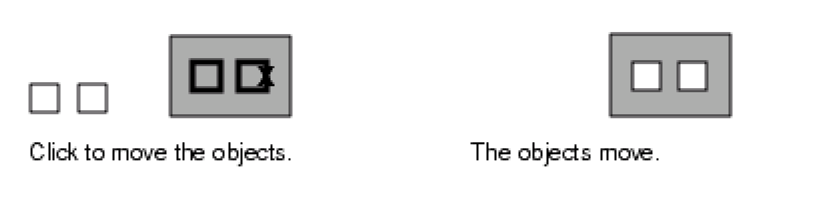
Click right.

Each time you **click right**, the instance or array rotates 90 degrees counterclockwise.

Move, copy

To move objects using the Move command,

1. Choose **Edit - Move [m].**
2. Select one or more objects.
3. **Click** where you want to move the objects



Copying Objects

1. Choose **Edit - Copy [c].**
2. Select one or more objects.

Click to select the first object, which point serves as a reference point for moving the copies.

Create a selection box, or press Shift while selecting all objects, and the Copy command prompts you for a reference point.

3. **Click** where you want to move the objects

Stretch

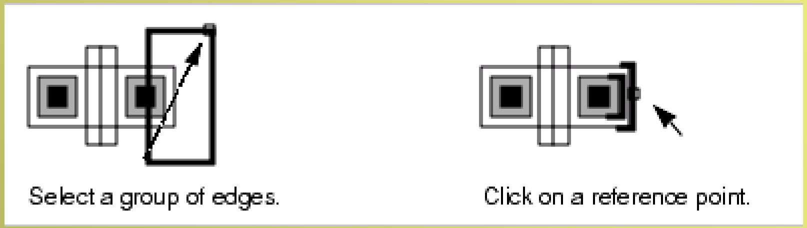
To stretch an object,

1.Choose **Edit - Stretch [s].**

2.Select the edges or corners you want to stretch.

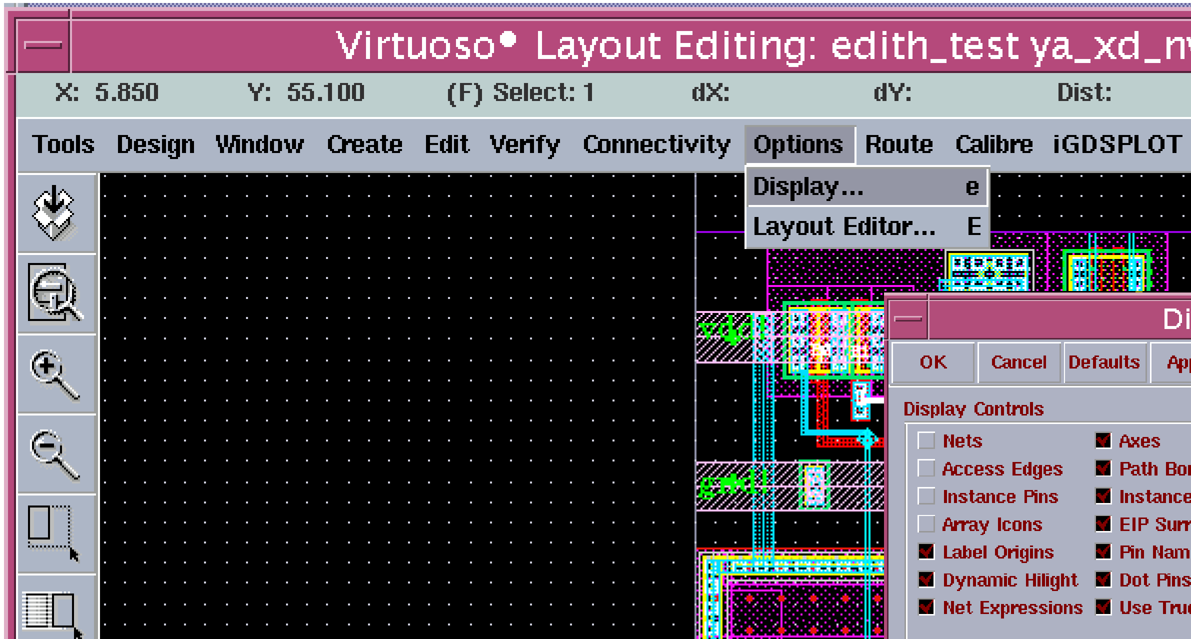
In partial selection mode, if you press **Shift** and **click** on an edge or corner, that edge or corner is the reference point for the stretch.

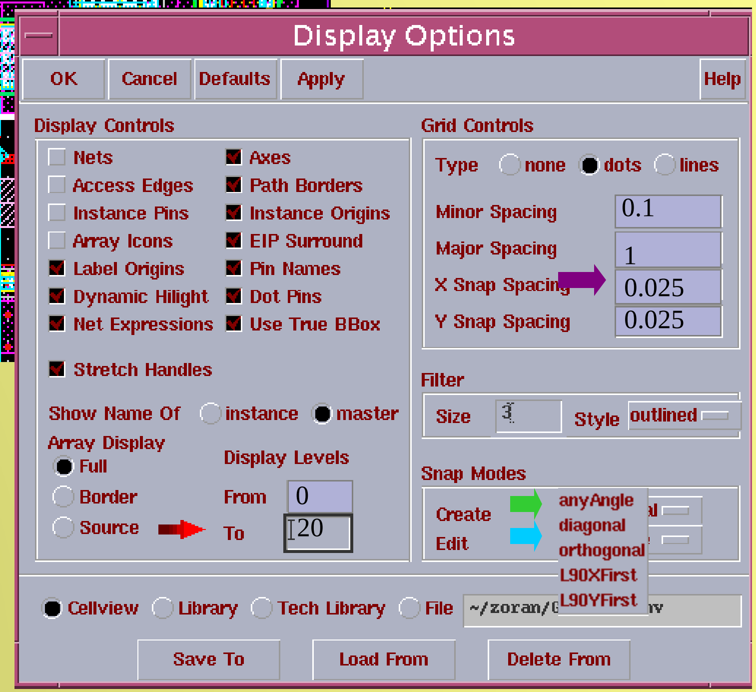
If you create a selection box around the edges, Stretch prompts you for a reference point. The reference point does not have to be on the selected shapes.



3 .Move the cursor and click to stretch the edges.

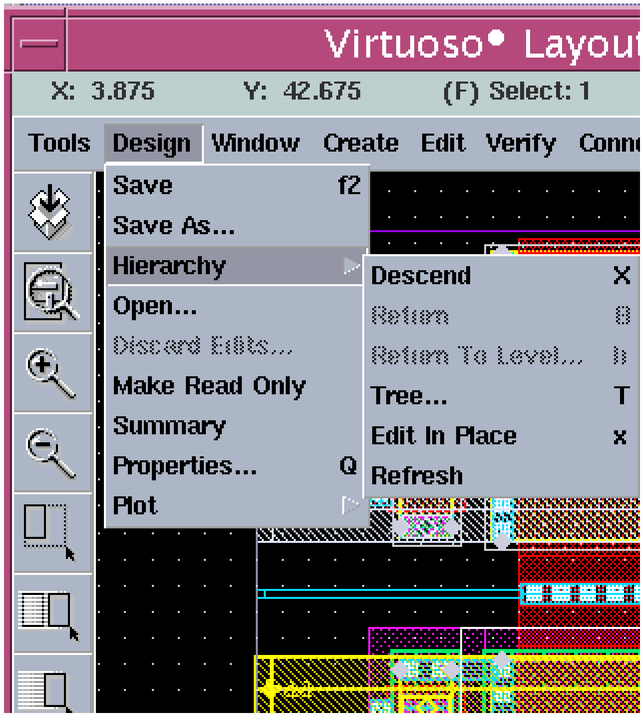
Display options editing



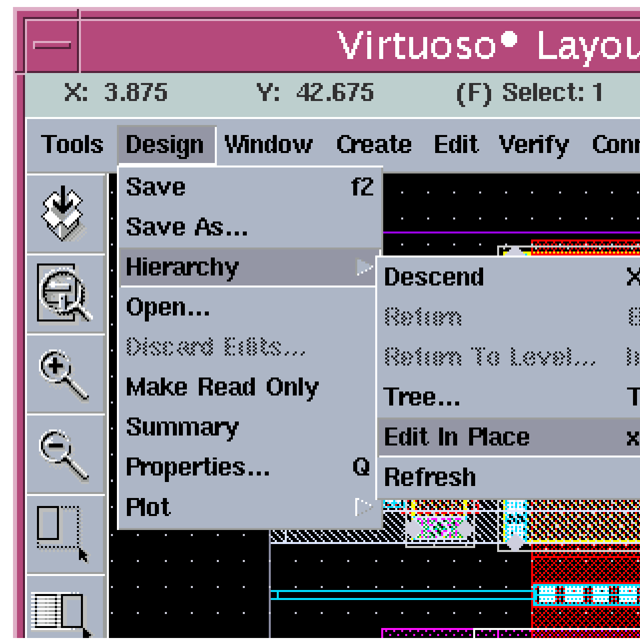


- **Insert DISPLAY levels**
- **Spacing**
- **Grid**
- **Create angle option**
- **Edit angle option**

Hierarchy



Go inside the instance

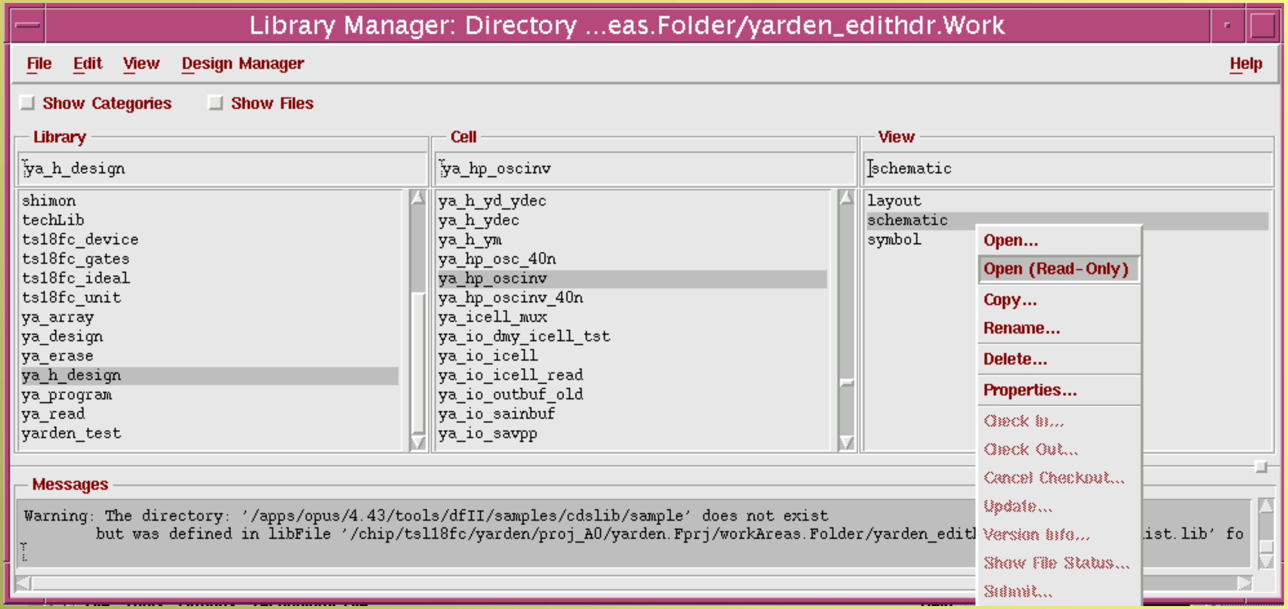


Go outside the instance

Open schematic view

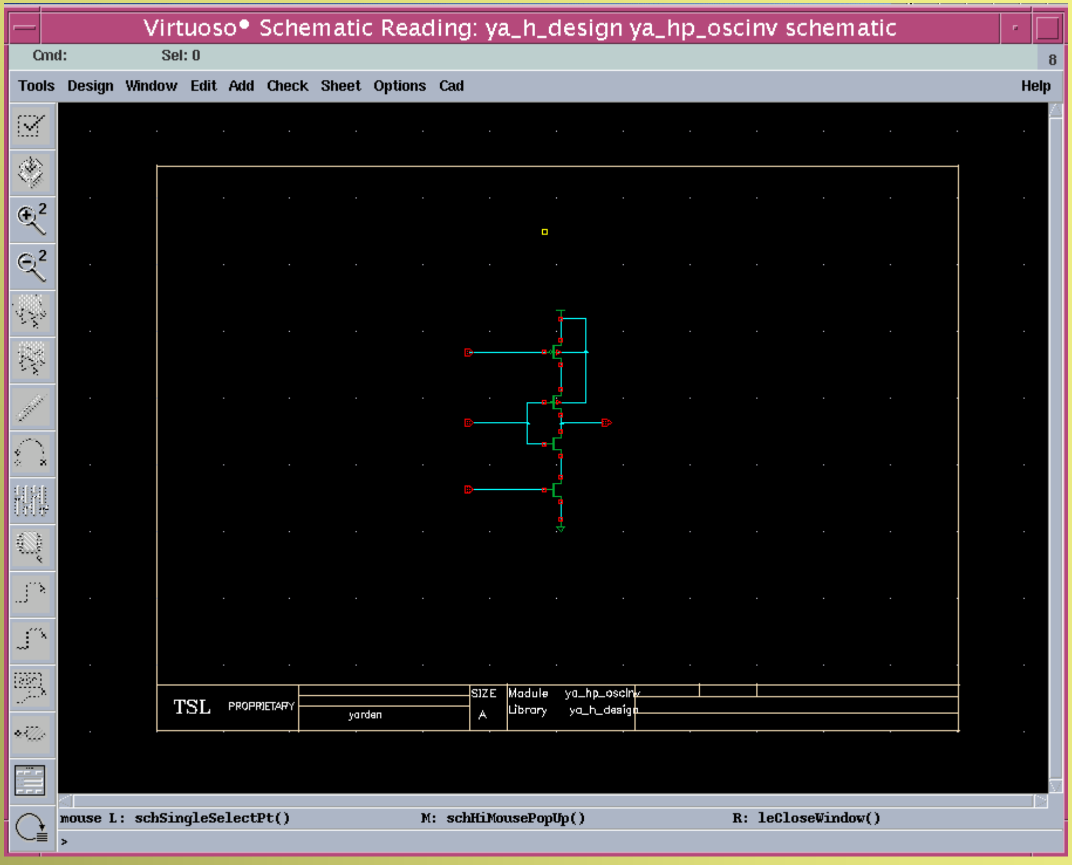
To open existing schematic view

on need cell click with right mouse pick and Open (read-only)



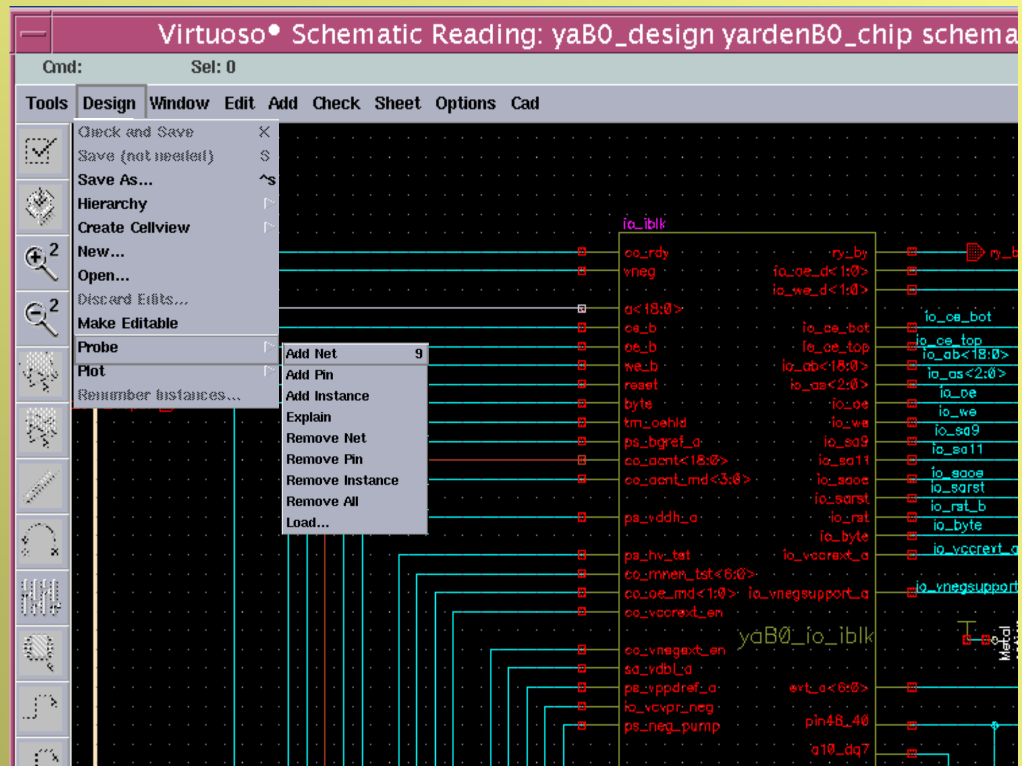
Schematic view

- All basic elements (pins, vdd, vss, devices)
- Are instances from existing libraries.



Schematic view signals probe

Design-> Probe -> Add Net

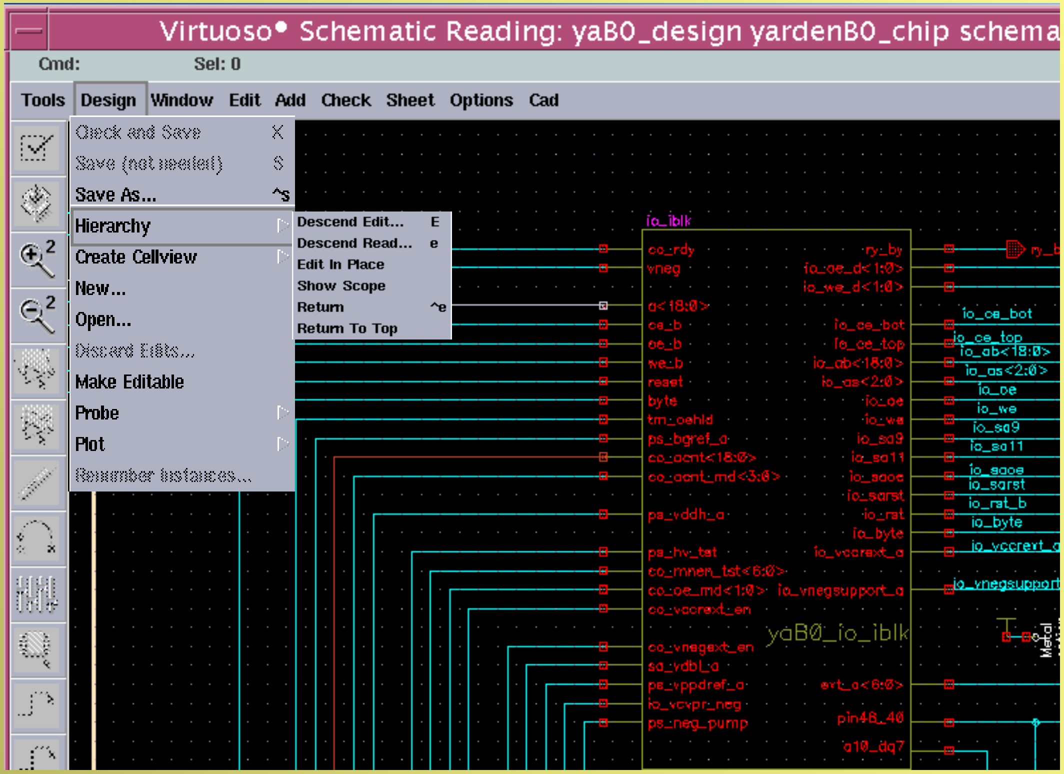


Schematic view hierarchy

Click on block

Design -> Hierarchy -> Descend Edit

Design -> Hierarchy -> Descend Read



Making Cell views Editable

To make a cell view editable

- Choose Design - Make Editable

You can change a cell so it is read only (you cannot edit it).

To make a cell view read only,

- Choose Design - Make Read Only

When the cell view is set to read only, this command is replaced by Design - Make Editable, which, when chosen, returns the cell view to edit mode. Design - Make Editable works only if the UNIX permissions on the file are set to enable write.